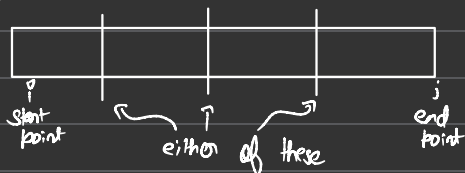


Partition OP $\rightarrow$ Tough pattern

$\rightarrow$ Solve a problem in a pattern.

eg: $(1 + 2 + 3) \times (5)$ or $(1 + 2) \times (3 \times 5)$



Matrix Chain Multiplication

ABC , $A = 10 \times 30$, $B = 30 \times 5$, $C = 5 \times 60$

$[]_{10 \times 30} \times []_{30 \times 5} \rightarrow \text{No. of operations required} = 10 \times 30 \times 5 = 1500$

$(A B C) = (10 \times 30 \times 5) + (10 \times 5 \times 60) = 1500 + 3000 = 4500 \text{ operations}$

$A (B C) = (30 \times 5 \times 60) + (10 \times 30 \times 60) = 9000 + 18000 = 27000 \text{ operations}$

Given n matrices return the minimum no of operations to multiply them $\rightarrow 4000$

Given : arr $[] \rightarrow \{10, 20, 30, 40, 50\}$ $N = 5$

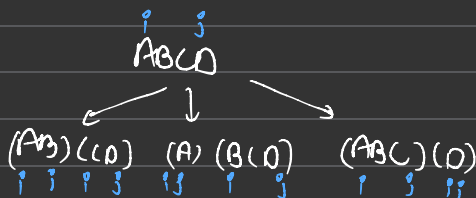
$\rightarrow$ dimension of $n-1$ matrix.

$A \rightarrow 10 \times 20$, $B \rightarrow 20 \times 30$, $C \rightarrow 30 \times 40$, $D \rightarrow 40 \times 50$

For i^{th} matrix $\rightarrow A[i-1] \times A[i]$

Rules for partition OP

1. Start with entire block/array. (i, j)
2. Try all partitions (run a loop to try all partitions)
3. Return the best possible two partitions.



$f(i, j) \rightarrow$ returns the minimum multiplications to multiply matrix $i \rightarrow$ matrix j

if $(i == j)$ return 0;

mini
= 1e9

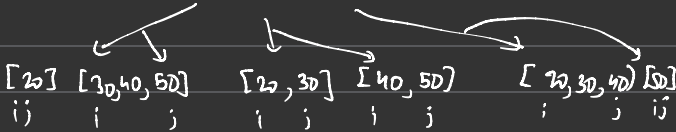
for $(k \rightarrow i \rightarrow j-1)$

$$\text{steps} = A[i-1] \times \text{arr}[k] \times \text{arr}[j] + f(i, k) + f(k+1, j);$$

$$\text{mini} = \min(\text{mini}, \text{steps});$$

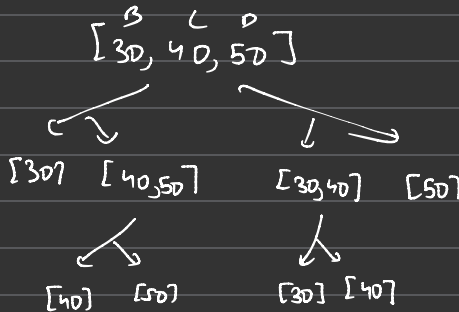
$$f(10, 20, 30, 40, 50)$$

A B C D



$$10 \times 20 \times 50$$

$$+ 0 + [30, 40, 50]$$



$$[10, 20, 30, 40, 50]$$

A B C D

$$k = (i \rightarrow j-1)$$

$$f(i, k), f(k+1, j)$$

$$A \times B \times C \times D$$

$$(10 \times 20) (20 \times 30) (30 \times 40) (40 \times 50)$$

$$(20 \times 40) (40 \times 50)$$

$$20 \times 50$$

$$\text{no of operations} = 10 \times 20 \times 50$$

Tabulation $\rightarrow$ dp[N][N]

1. Copy the base case
2. Write the changing states (i, j) $(i = n-1 \rightarrow 0)$ $(j = i+1 \rightarrow n-1)$
3. Copy the recurrence